EDA on SALARY DATA

CODE:

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder, StandardScaler

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

df = pd.read_csv("Salary Data.csv")

df.head()

	Age	Gender	Education Level	Job Title	Years of Experience	Salary
0	32.0	Male	Bachelor's	Software Engineer	5.0	90000.0
1	28.0	Female	Master's	Data Analyst	3.0	65000.0
2	45.0	Male	PhD	Senior Manager	15.0	150000.0
3	36.0	Female	Bachelor's	Sales Associate	7.0	60000.0
4	52.0	Male	Master's	Director	20.0	200000.0

df.shape

df.info()

df.describe()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 375 entries, 0 to 374
Data columns (total 6 columns):
 #   Column               Non-Null Count  Dtype  
---  ------               --------------  -----  
 0   Age                  373 non-null    float64
 1   Gender               373 non-null    object 
 2   Education Level      373 non-null    object 
 3   Job Title            373 non-null    object 
 4   Years of Experience  373 non-null    float64
 5   Salary               373 non-null    float64
dtypes: float64(3), object(3)
memory usage: 17.7+ KB
```

	Age	Years of Experience	Salary
count	373.000000	373.000000	373.000000
mean	37.431635	10.030831	100577.345845
std	7.069073	6.557007	48240.013482
min	23.000000	0.000000	350.000000
25%	31.000000	4.000000	55000.000000
50%	36.000000	9.000000	95000.000000
75%	44.000000	15.000000	140000.000000
max	53.000000	25.000000	250000.000000

df.isnull().sum()

```
Age                    2
Gender                 2
Education Level        2
Job Title              2
Years of Experience    2
Salary                 2
dtype: int64
```

df["Age"].fillna(df["Age"].mean())

df["Years of Experience"].fillna(df["Years of Experience"].mean())

df["Gender"].fillna(df["Gender"].mode()[0])

df["Education Level"].fillna(df["Education Level"].mode()[0])

df["Job Title"].fillna(df["Job Title"].mode()[0])

df = df.dropna(subset=["Salary"])

df.isnull().sum()

```
Age                    0
Gender                 0
Education Level        0
Job Title              0
Years of Experience    0
Salary                 0
dtype: int64
```

le = LabelEncoder()

df['Gender'] = le.fit_transform(df['Gender'])

df['Education Level'] = le.fit_transform(df['Education Level'])

df['Job Title'] = le.fit_transform(df['Job Title'])

df.head()

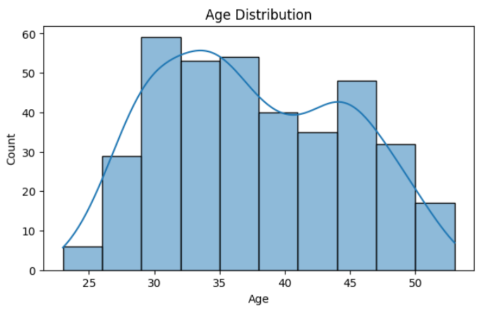
	Age	Gender	Education Level	Job Title	Years of Experience	Salary
0	32.0	1	0	159	5.0	90000.0
1	28.0	0	1	17	3.0	65000.0
2	45.0	1	2	130	15.0	150000.0
3	36.0	0	0	101	7.0	60000.0
4	52.0	1	1	22	20.0	200000.0

plt.figure(figsize=(7,4))

sns.histplot(df['Age'], kde=True)

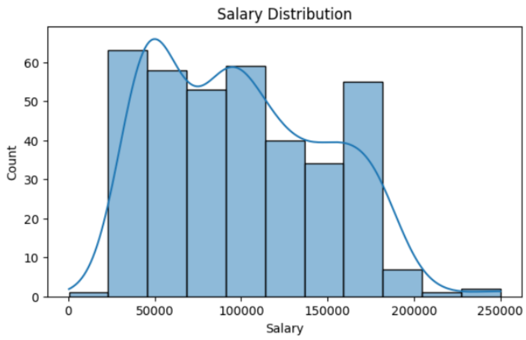
plt.title("Age Distribution")

plt.show()

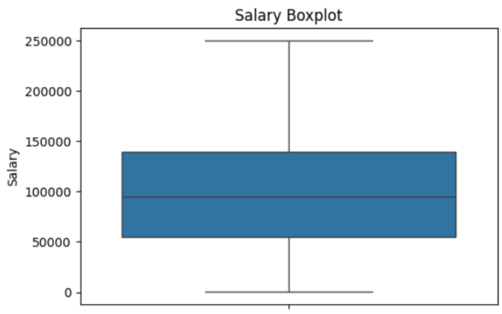


plt.figure(figsize=(7,4))

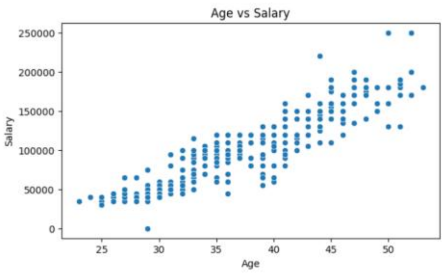
```
sns.histplot(df['Salary'], kde=True)
plt.title("Salary Distribution")
plt.show()
```




```
plt.figure(figsize=(6,4))
sns.boxplot(y=df['Salary'])
plt.title("Salary Boxplot")
plt.show()
```

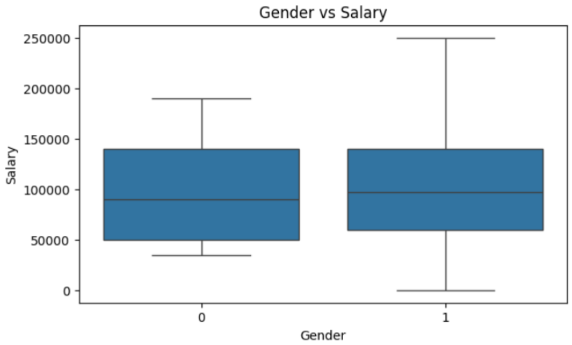



```
plt.figure(figsize=(7,4))
sns.scatterplot(x='Age', y='Salary', data=df)
plt.title("Age vs Salary")
plt.show()
```

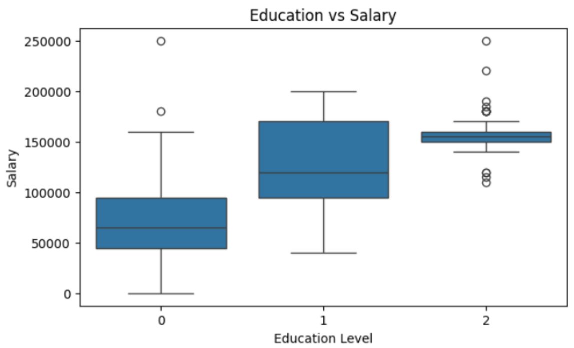



```
plt.figure(figsize=(7,4))
sns.scatterplot(x='Years of Experience', y='Salary', data=df)
plt.title("Experience vs Salary")
plt.show()
```

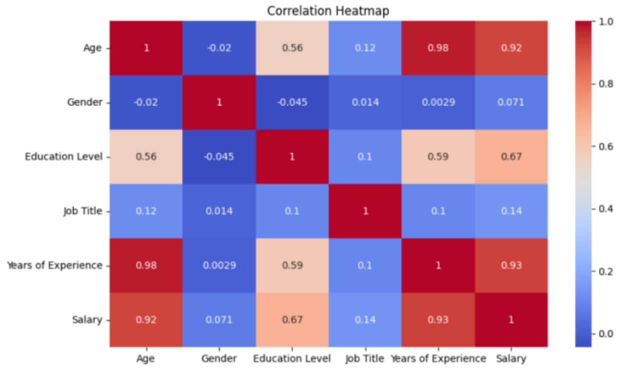

```
plt.figure(figsize=(7,4))
sns.boxplot(x='Gender', y='Salary', data=df)
plt.title("Gender vs Salary")
plt.show()
```




```
plt.figure(figsize=(7,4))
sns.boxplot(x='Education Level', y='Salary', data=df)
plt.title("Education vs Salary")
plt.show()
```

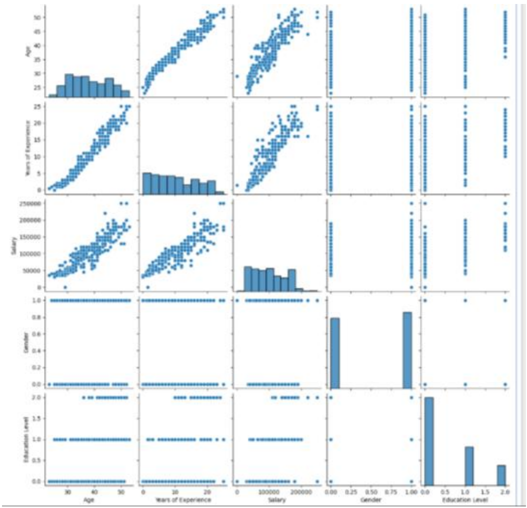



```
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



sns.pairplot(df[['Age','Years of Experience','Salary','Gender','Education Level']])

plt.show()



X = df[['Age','Gender','Education Level','Years of Experience']]

y = df['Salary']

X_train, X_test, y_train, y_test = train_test_split(

X, y, test_size=0.2, random_state=42)

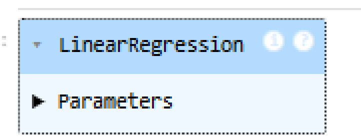
scaler = StandardScaler()

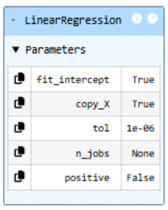
X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)

model = LinearRegression()

model.fit(X_train, y_train)





y_pred = model.predict(X_test)

print("MAE :", mean_absolute_error(y_test, y_pred))

print("MSE :", mean_squared_error(y_test, y_pred))

print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))

print("R² :", r2_score(y_test, y_pred))

```
MAE : 10613.910121260109
MSE : 232595028.5296489
RMSE: 15251.06647187825
R² : 0.9029876738433329
```

from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure(figsize=(8,6))

ax = fig.add_subplot(111, projection='3d')

ax.scatter(df['Age'], df['Years of Experience'], df['Salary'])

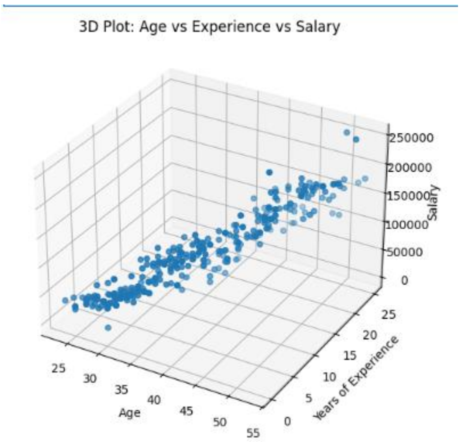
ax.set_xlabel("Age")

ax.set_ylabel("Years of Experience")

ax.set_zlabel("Salary")

ax.set_title("3D Plot: Age vs Experience vs Salary")

plt.show()



pd.DataFrame(model.coef_, X.columns, columns=['Coefficient'])

	Coefficient
Age	24616.094533
Gender	3802.940602
Education Level	10728.963626
Years of Experience	13648.159540